



Creaky voice facilitates more efficient phonological processing of Mandarin Tone 3

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Abstract

Though tone recognition is primarily influenced by pitch cues, phonation type may also play a role. For example, creaky voice, a non-modal phonation, can enhance Mandarin Tone 3 (T3) recognition. While most studies explore creaky voice in tone recognition behaviorally, its neural processing remains unclear. This study aims to investigate tone processing using electroencephalography to examine how the brain integrates pitch and creaky voice to process T3. We measured Mismatch Negativity (MMN) responses of 10 native Mandarin speakers exposed to an oddball paradigm with real-word single syllable stimuli in two conditions: creak and non-creak. Results showed stronger MMN responses in the creak condition, with larger mean MMN amplitude, shorter peak MMN latency, and larger negative peak amplitude. These findings suggest that creaky voice enhances T3 distinctiveness and facilitates more efficient phonological processing, contributing to a deeper understanding of tone processing mechanisms.

Index Terms: Tone recognition, phonation, creaky voice, Mandarin tone

1. Introduction

While pitch is the dominant cue in lexical tone recognition [1, 2], other factors, such as phonation type [3, 4], also play a role. Phonation refers to the sound generated as the vocal folds vibrate during speech production. Variation in the way the vocal folds vibrate leads to differences in phonation that can be generally categorized into two types: modal phonation and non-modal phonation. Modal phonation, the most common and default voice quality, is characterized by regular, periodic vibrations of the vocal folds. In contrast, non-modal phonation includes various complex patterns of vocal fold vibration that differ from modal phonation, including irregular or aperiodic vibration [5]. Creaky voice, a type of non-modal phonation, is often marked by this irregular vibration [6]. It typically co-occurs with specific tones and serves as an additional cue in tone recognition, even in tonal languages without phonological phonation contrasts, such as Mandarin [3, 7, 8, 9]. Despite its relevance, most research on tone recognition has focused on pitch-based cues, with limited attention to the role of phonation [1, 10, 11]. This study aims to provide deeper insights into the mechanisms underlying tone recognition by examining how tone is processed when non-modal phonation cues such as creaky voice are introduced.

As a tone language, Mandarin has four distinct tones: Tone 1 (T1) is a high-level tone, Tone 2 (T2) is a rising tone, Tone 3 (T3) is a low-dipping tone, and Tone 4 (T4) is a falling tone [12]. Among the four tones, creaky voice can occur at the lowest portion of T2, T3, and T4, specifically, at the onset of T2, the turning point of T3, and the offset of T4 [8]. Notably, creaky voice is most frequently associated with T3. Since Mandarin tones are primarily distinguished by pitch, phonation contrasts are not a defining feature. Consequently, phonological studies suggest that creaky voice functions merely as a redundant cue in Mandarin tone [7]. However, perception studies indicate that creaky voice enhances both the speed and accuracy of T3 recognition, while it does not provide similar benefits for other tones, regardless of whether natural or resynthesized stimuli are used. For instance, Belotel-Grenié and Grenié [13] found that listeners recognized T3 more quickly when it was realized with creaky voice in a gating task. Similarly, Huang [3] superimposed tone contours for each of the Mandarin tones T1-T4 onto T1 syllables and added creaky voice to the resynthesized stimuli, reporting that creaky voice improved T3 recognition but hindered recognition of T1 and T4.

Moreover, distinguishing T3 from T2 based solely on pitch can be challenging due to the acoustic similarity of their pitch contours [14, 15]. However, research on the role of phonation cues on Mandarin tone perception suggests that listeners rely more heavily on creaky voice for recognition of T3. Notably, Yang [16] found that T3 was more easily recognized in a tone continuum generated from a T3 syllable than from other-tone syllables, provided that syllables of all four tones maintained their original phonation cues, specifically, the T3 syllable retained creaky voice, and other-tone syllables maintained modal phonation cues. Likewise, Way [17] found that T2 was more likely to be perceived as T3 when creakiness was more pronounced in the middle portion of T2, leading to higher incorrect recognition of T2. Given these findings, creaky voice would seem not to be a redundant cue but rather a useful cue for T3 recognition.

Little is known about the neural processing of creaky voice in tone recognition. Most studies on Mandarin tone processing have focused on pitch cues. For instance, Chandrasekaran [18] used an oddball paradigm in an electroencephalographic (EEG) study to investigate Mandarin tone processing of pitch dimensions (e.g., pitch height and pitch contour), finding that T3 was more difficult to distinguish from T2 than from T1 when relying solely on pitch cues. Similarly, Li and Chen [19] suggested that pitch alone may be insufficient for reliably differentiating T3 from T2. However, despite behavioral evidence highlighting the importance of creaky voice in T3

discrimination, it remains unclear how the brain integrates pitch and creaky voice during T3 processing.

Mismatch Negativity (MMN) has been widely used in speech research to reflect the neural response to changes in acoustic cues [20]. Three key measures are commonly used to evaluate the robustness and efficiency of MMN response: (1) mean MMN amplitude, which reflects the overall strength of a neural response; (2) peak MMN latency, which indicates the timing of deviance detection; and (3) peak MMN amplitude, which provides insights into the efficiency of neural processing of stimulus deviations. In general, as the magnitude of stimulus deviation increases, neural activation intensifies, leading to a larger mean MMN amplitude, a shorter peak MMN latency, and a larger negative peak MMN amplitude [20]. In addition, the MMN scalp distribution can offer insights into which types of information are processed: the right hemisphere is typically associated with acoustic information encoding, while the left hemisphere is more specialized for encoding linguistic information [21, 22].

In this study, we measured MMN responses to T3 with and without creaky voice, aiming to answer the following questions: (a) How does the brain integrate pitch and creaky voice to distinguish T3 from T2? (b) At which linguistic level, acoustic or phonological, does creaky voice facilitate T3 processing? We predicted that (1) the presence of creaky would enhance T3 distinctiveness, evoking an earlier and stronger MMN; (2) Creaky voice would enhance phonological encoding at the early stage of T3 processing, showing an early bilateral scalp distribution of MMN responses.

2. Methods

2.1. Participants

Ten native Mandarin-speaking participants (20-28 years old; 8 females, 2 males) currently living in Australia took part in the experiment. All participants were born and raised in China and were either university students or had obtained a bachelor's degree. They were all right-handed and had normal hearing thresholds between 250 Hz and 8000 Hz. None reported any language difficulties, neurological impairments, or a history of serious head injury. The study was approved by Macquarie University's Human Ethics Committee. All participants provided written informed consent before the experiment began and received compensation upon completion. The experiment was conducted in the EEG lab at Macquarie University.

2.2. Design

In this study, we used the T2-T3 contrast, in which T3 had two variants: creak and non-creak. We employed the passive oddball paradigm which is widely used to elicit MMN. In this paradigm, a sequence of repetitive standard stimuli is occasionally interrupted by a rare deviant stimulus. MMN is triggered by the brain's detection of the changes introduced by the deviant relative to the standard. We adopted this approach to measure MMNs in two conditions: a creak condition and a non-creak condition. In the creak condition, creak T3 served as the deviant stimulus, while non-creak T2 acted as the standard stimulus. In the non-creak condition, non-creak T3 was the deviant, with non-creak T2 again as the standard. Each condition included 700 trials, comprising 600 standard stimuli (86%) and 100 deviant stimuli (14%). EEG was used to collect data due to its high temporal resolution, enabling the capture of

neural activity on a millisecond scale essential for studying rapid auditory processing.

2.3. Stimuli

The target stimuli consisted of two monosyllabic CV Mandarin words distinguished only by lexical tones: /ja2/ ('tooth') and /ja3/ ('dumb'). For /ja3/ ('dumb'), there were two variants: creak /ja3/ and non-creak /ja3/. All stimuli were resynthesized.

2.3.1. Tone resynthesis

Three monosyllabic CV words /ma/, /ja/, and /wa/ produced with all four tones were used to fit the tone contours. The target tone contours were T2 and T3, while T1 and T4 contours were used to ensure proper distribution of T2 and T3 within the speaker's natural pitch range. A native Mandarin male speaker (23 years old) uttered each word in isolation. Each word was repeated 15 times. Recordings were made using TotalMix FX audio processing software with a high-quality Neumann TLM103 microphone at a sampling rate of 44,100 Hz in a sound-attenuated room at Macquarie University Phonetics Lab.

Each tone category consisted of 45 tokens (3 words * 15 repetitions). For each token, thirteen F0 points were extracted from its pitch track at equal time intervals using Praat [23]. Based on the F0 values, four tone contours were fitted using second-order polynomial models in R [24], as shown in Equation 1 to 4. The fitted T2 and T3 contours were then applied to the word /ja1/ to create /ja2/ and /ja3/ using the pitch synchronous overlap-add (PSOLA) method in Praat [23].

$$T1: y = 134.03 + 3.26x - 0.17x^2$$

Equation 1: *The second-order polynomial models of T1*

$$T2: y = 142.55 - 8.39x + 0.48x^2$$

Equation 2: *The second-order polynomial models of T2*

$$T3: y = 136.22 - 6.4x + 0.55x^2$$

Equation 3: *The second-order polynomial models of T3*

$$T4: y = 141.67 + 8.68x - 0.75x^2$$

Equation 4: *The second-order polynomial models of T4*

2.3.2. Creak resynthesis

The creak /ja3/ pitch contour was derived from the resynthesized non-creak /ja3/ by introducing irregular F0 serving as the primary cue for a jittered "creak" portion using Praat [23]. F0 values in the creak portion ranged from 60 Hz to 105 Hz. Specifically, an irregular F0 was inserted at 33.3% of the total syllable duration, covering approximately one-third of the entire syllable [3].

All stimuli were rated by native Mandarin-speaking listeners to confirm their naturalness. Of these, the recognition accuracy of /ja2/ was 97.5%, /ja3/ was 79.2%, and creak /ja3/ was 100%. The duration of all the stimuli was 400 ms, and the amplitude was scaled to 70 dB.

2.4. Procedure

Participants watched a self-selected movie with the sound muted while passively listening to the auditory stimuli presented at 70 dB via a loudspeaker. They were seated on a sofa approximately 1 meter from the screen and 0.8 meters from the loudspeaker. To minimize distractions, they were instructed to avoid extraneous movements. Chinese subtitles were provided to help maintain their focus on the movie. At the end

of the experiment, they were required to provide a brief summary of the movie to confirm their engagement.

The order of conditions was counterbalanced across participants. Within each condition, stimuli were presented in a pseudorandomized sequence, ensuring that each deviant stimulus was preceded by at least two standard stimuli. The inter-trial interval ranged from 600 to 800 ms, and each condition began with 10 extra standard stimuli.

2.5. EEG data acquisition and analysis

The EEG signals were recorded via a SynAmps 2 amplifier (NeuroScan Inc.) with a 64-channel EASYCAP (EASYCAP GmbH, Germany) carrying Ag/AgCl electrodes placed according to the international 10–20 system. Vertical eye blinks and horizontal eye movements were monitored using electrodes placed above and below the right eye (VEOG) and at the outer canthi of both eyes (HEOG). The EEG was recorded with an online reference to the left mastoid. All electrode impedance levels were kept below 5 k Ω . The signals were continuously recorded and digitized at a 24-bit resolution with a sampling rate of 1000 Hz, using a bandpass filter of 0.05 Hz to 400 Hz.

Pre-processing of EEG signals was implemented using fieldtrip [25]. The raw EEG data were first re-referenced to the mastoids and then filtered using a 0.5 – 30 Hz band-pass filter. Eye movement and blink artifacts were corrected using independent component analysis (ICA). The corrected data were segmented into epochs from -200 ms to 600 ms relative to the word onset. For each epoch, there was a time window of 200 ms preceding the onset of the words serving as baseline correction. Epochs with amplitudes exceeding ± 75 μ V or ocular artifacts were excluded from further analyses. The mean acceptance rate of epochs was above 80% in each condition.

MMN was calculated by subtracting the standard event-related potential (ERP) waveform from the deviant ERP waveform. MMN was identified through visual inspection of the grand-average MMN and its corresponding topographic distribution maps. Additionally, 15 fronto-central electrodes, where MMN amplitudes were expected to peak, were chosen for analysis: F1, F3, FC1, FC3, C1 and C3 on the left hemisphere, Fz, FCz and Cz located along the midline, and F2, F4, FC2, FC4, C2 and C4 on the right hemisphere. The grand-average MMN refers to the averaged MMN waveform across all electrodes of interest and all participants. Their corresponding topographic maps were plotted in 20ms intervals, with the last subfigure plotted according to the actual residual duration. Mean MMN amplitude was calculated by averaging the ERP signal across trials at an electrode within a specific time window. Peak MMN latency was defined as the time point of the largest negative peak of an electrode within the time window, while peak MMN amplitude was the corresponding voltage at this peak.

Linear mixed-effect models were performed to evaluate mean MMN amplitude, peak MMN latency, and peak MMN amplitude across conditions using the lmerTest package [26] in R [24]. All models were fitted with maximal random structures permissible by the data without singularity or convergence issues and were subsequently simplified and selected based on a likelihood ratio test to identify the best-fitting model for each analysis [27] (see below for details of final model structure). If a significant main effect of a multilevel factor or a significant interaction was observed, post hoc pairwise comparisons were conducted using the emmeans package [28].

3. Results

3.1. The existence of MMN

The grand-average MMNs are shown in Figure 1. Their topographic distribution maps in the creak and non-creak conditions are presented in Figure 2 and Figure 3, respectively.

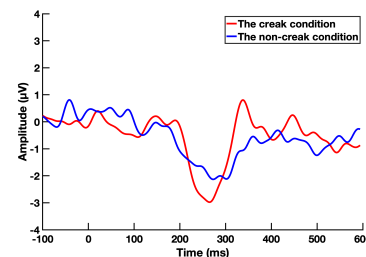


Figure 1: The grand-average MMNs in the creak and non-creak conditions

Based on the grand-average MMNs and corresponding topographic distribution maps, clear MMN responses were observed within specific time windows: approximately 229–289 ms in the creak condition and 252–312 ms in the non-creak condition. Visually, MMN appeared stronger in the creak condition compared to the non-creak condition. Additionally, MMN elicited in the creak condition showed bilateral scalp distribution during the early processing stage (229–249 ms), while MMN in the non-creak condition exhibited right-hemisphere dominance (252–302 ms).

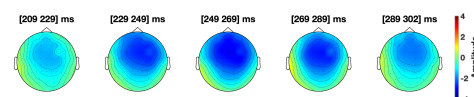


Figure 2: Topographic distribution maps of the difference waveforms in the creak condition (209–302 ms)

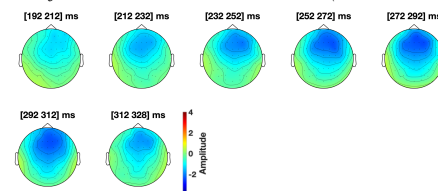


Figure 3: Topographic distribution maps of the difference waveforms in the non-creak condition (192–328 ms)

3.2. Mean MMN amplitude

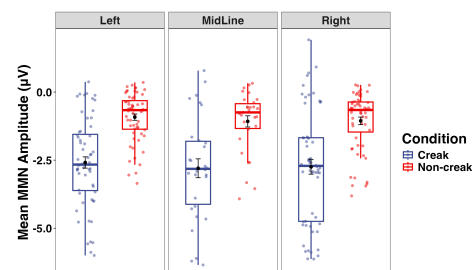


Figure 4: Mean MMN amplitudes in the creak and non-creak conditions

Figure 4 shows mean MMN amplitudes in the creak and non-creak conditions. A linear mixed-effects model was fitted with Condition (Creak and Non-creak) and BrainArea (Left, Right, and MidLine) as fixed effects, and Subject as a random

intercept. The results revealed a significant main effect of Condition, $F(1, 285) = 113.74, p < .001$. No other significant main effects or interactions were found. These results indicate that mean MMN amplitude in the creak condition ($M = -2.70 \mu V, SD = 1.85 \mu V$) was significantly more negative than that in the non-creak condition ($M = -1.00 \mu V, SD = 1.02 \mu V$).

3.3. Peak MMN latency

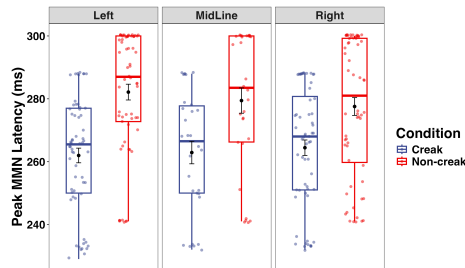


Figure 5: Peak MMN latencies in the creak and non-creak conditions

Figure 5 shows MMN peak latencies in the creak and non-creak conditions. A linear mixed-effect model was applied with Condition (Creak and Non-creak), and BrainArea (Left, Right, and MidLine) as fixed effects, and Subject as a random intercept. The results revealed a significant main effect of Condition, $F(1, 285) = 91.41, p < .001$. No other significant main effects or interactions were found. These results suggest that MMN peaked significantly earlier in the creak condition ($M = 263.14 \text{ ms}, SD = 18.69 \text{ ms}$) than in the non-creak condition ($M = 279.76 \text{ ms}, SD = 20.95 \text{ ms}$).

3.4. Peak MMN amplitude

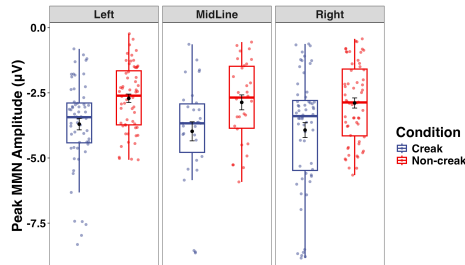


Figure 6: Mean peak amplitudes of MMN in the creak and non-creak conditions

Figure 6 shows peak MMN amplitudes in the creak and non-creak conditions. A linear mixed-effect model was fitted with Condition (Creak and Non-creak) and BrainArea (Left, Right, and MidLine) as fixed effects, and Subject as a random intercept. The results revealed a significant main effect of Condition, $F(1, 285) = 34.96, p < .001$. No other significant main effects or interactions were found. These results suggest that peak MMN amplitude was more negative in the creak condition ($M = -3.85 \mu V, SD = 1.98 \mu V$) compared to the non-creak condition ($M = -2.81 \mu V, SD = 1.42 \mu V$).

4. Discussion

The current study investigated how the brain integrates pitch and creaky voice to distinguish T3 from T2. Our findings supported the predictions that creaky voice would enhance the distinctiveness of T3, leading to stronger neural responses and more efficient phonological processing. Specifically, we

observed a larger mean MMN amplitude, a shorter peak MMN latency, and a larger negative peak MMN amplitude in the creak condition, along with a bilateral scalp distribution during the earlier stage of processing. These results suggest that creaky voice facilitates T3 processing by enhancing its distinctiveness, leading to more efficient phonological encoding.

Our results offer several important theoretical implications. First, they provide further neural evidence that creaky voice is a useful cue for T3 processing supporting behavioural studies [3, 13, 16]. Creak T3 elicited both a larger and faster MMN compared to non-creak T3, highlighting that creaky voice enhances T3 distinctiveness and speeds up neural processing. A larger negative peak MMN amplitude evoked by creak T3 further indicates the enhancement of efficiency in T3 processing, supporting previous studies that the larger the stimulus deviation, the more robust and efficient the processing [20].

Second, creak T3 may engage both acoustic and phonological encoding early on, whereas non-creak T3 primarily involves acoustic processing. MMN topography showed a bilateral distribution in the creak condition during early processing, while the non-creak condition showed right-hemisphere dominance. These findings suggest that creaky voice may facilitate access to phonological representations during tone processing. These interpretations are consistent with neuroimaging findings that the right hemisphere processes general acoustic information, such as pitch, while the left hemisphere specializes in linguistic processing, including phonological processing [21]. However, since the current statistical analysis did not include processing time as a factor, these interpretations remain tentative. Future research should adopt time-sensitive methods to confirm that these neural processing differences are distinct in brain areas.

Third, our findings challenge a purely pitch-centric view of tone processing by emphasizing the significant role of creaky voice in tone recognition. The current study highlights the importance of incorporating non-pitch cues into our understanding of tone recognition, particularly in languages where pitch is traditionally considered to be the primary cue.

Finally, while the current study provides valuable insights into the contribution of creaky voice to tone processing, further research is needed to determine whether creaky voice truly enhances phonological encoding, particularly when accounting for language experience. Additionally, further studies should investigate how creaky voice functions in non-word contexts, where linguistic experience has less influence. This would help clarify the role of creaky voice in tone processing in the absence of familiar linguistic context.

5. Conclusions

This study investigated how creaky voice contributes to Mandarin T3 processing. The results show that creaky voice enhances the distinctiveness of T3, eliciting a stronger MMN with a larger mean MMN amplitude, an earlier MMN peak, a more negative peak MMN amplitude, and a bilateral distribution in the early stages. These preliminary findings suggest that creaky voice facilitates more effective phonological coding in Mandarin tone processing. Furthermore, they emphasize the importance of considering non-pitch cues, such as phonation type, in tone processing, contributing to a more comprehensive understanding of the cognitive mechanisms underlying tone recognition.

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7. References

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